

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

Mid Semester Exam October 2025

Program:- MCA (III Semester)

Course:- **Theoretical Computer Science (CS-414)**

Max Marks:- 20

Time:- 90 Minutes

Date: 06-10-2025

Timing: 09:15AM-10:45AM

Note: Attempt all the questions, assume suitable data if missing.

Q. 1. Let $\Sigma = \{0, 1\}$, Language $L = \{ W \mid W \text{ contains the numbers } 1, 2, 4, 8, \dots, 2^n, \dots \text{ written in binary, } n \geq 0 \}$, Design the Deterministic Finite Automata and write the Regular expression for the above language. [2]

Q. 2. For the given Finite Automate shown in the Figure 1, Find the Language and Regular Expression. [2]

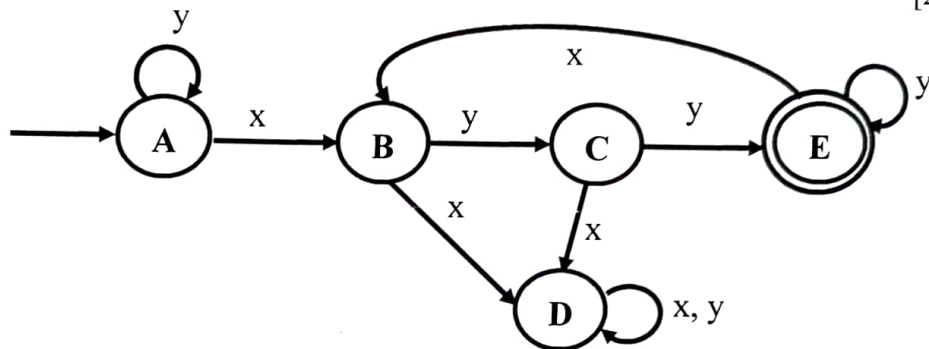


Figure 1

Q. 3. The Finite state machine shown in the Figure 2 accepts binary strings with combinations of 0's and 1's. Find the Language and Regular Expression for the same machine. [2]

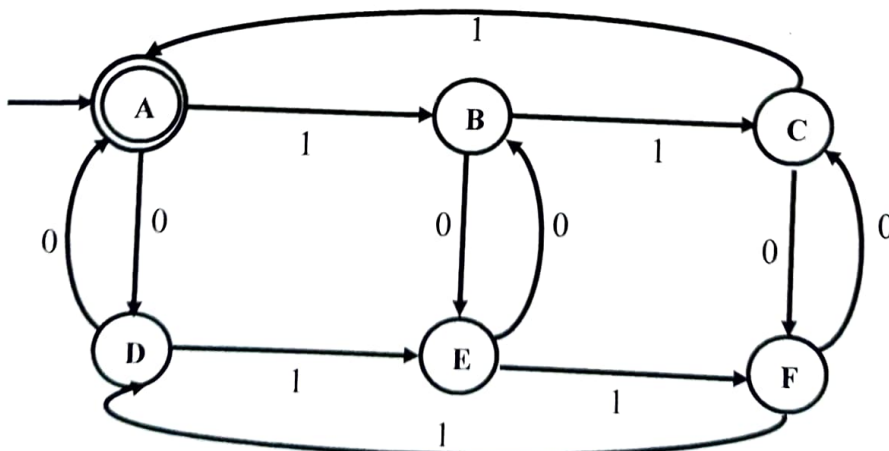


Figure 2

Q. 4. Let $\Sigma = \{a, b\}$, Language $L = \{W \mid W \text{ contains all odd length palindromes}\}$, Write the Grammar for the above language using Derivation Tree and verify it with suitable example. [2]

Q. 5. Check the following statements and verify whether these are True/False with proper justification: [4]

- The language $L = \{a^n b^n \mid n \geq 0 \text{ with } \Sigma = \{a, b\}\}$ is regular.
- The language $L = \{a^n \mid n \text{ is a prime with } \Sigma = \{a\}\}$ is regular.
- The language $L = \{w \mid w \text{ has } 2k+1 \text{ b's for some } k \in \mathbb{N} \text{ with } \Sigma = \{a, b\}\}$ is regular.
- The language $L = \{w \mid w \in \Sigma^* \text{ with } \Sigma = \{0, 1\}\}$ is regular.

Q. 6. For following Regular Grammar:

$$S \rightarrow A a B$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

$$B \rightarrow a \mid b$$

Find the NFA, Regular Language and Regular Expression. [3]

Q. 7. Define the Regular Grammar, DFA, NFA and also show the relationship between these three terminologies. [3]

Q. 8. Discuss the different properties of the regular expression with suitable example. [2]

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End Semester Exam December 2025

Program:- MCA (III Semester)

Course:- Theoretical Computer Science (CS414)

Max Marks:- 50

Time:- 180 Minutes

Exam date: 11-12-2025

Student Roll No.....

Note: Attempt all the questions, assume suitable data if missing.

Q.1. Let $\Sigma = \{0, 1\}$, Language $L = \{ W \mid W \text{ contains the string that starting with 01 and ending with 10} \}$, Design the DFA, NFA and Regular Expression for the above language.

[3+1+1=5]

Q.2. Let $\Sigma = \{a, b\}$, Language $L = \{ W \mid W \text{ contains string whose first two symbols are same} \}$, Design the minimal DFA, minimal NFA and Regular Expression for the above language.

[3+1+1=5]

Q.3. Let $\Sigma = \{a, b\}$, Language $L = \{ W \mid W \text{ contains a string of length at least 3} \}$. Write regular and Non regular Grammar for the above language.

[3]

Q.4. Write regular or non-regular Grammar for the following languages:

[2+2+2= 6]

a) $a^p b^p c^q d^q e^r f^r$, where $p, q, r \geq 1$

b) $a^m b^n c^n d^m$, where $m, n \geq 1$

c) $a^* + b^+$

Q.5. Construct a Moore machine that prints 1 for every occurrence 'abb'.

[3]

Q.6. Discuss properties of the regular language, and show the relationship between following languages: RL, CFL, CSL, DCFL.

[5]

Q.7. Write short notes on following:

[10]

a) Pumping Lemma.

b) Kleene Theorem.

c) Church's Thesis.

d) Mealy machine.

e) ϵ -NFA with suitable example.

Q.8. Construct PDA for the following language

[2+2=4]

a) Simple Palindrome (xcx^r) example of string aca.

b) $a^n b^{m+n} c^m$

Q.9. Define the role of the Turing machine, in Theoretical computer science. Design a Turing machine for following language Let $\Sigma = \{a, b\}$, Language $L = \{ \{a, b\}^* \{ab\} \{a, b\}^* \cup \{a, b\}^* \{ba\} \}$.

[2+3=5]

Q.10. In each case below, check whether given statement is a tautology, a contradiction, or neither, with proper justification:

[4]

a) $p \wedge \neg(p \rightarrow p)$

b) $(p \rightarrow \neg p) \vee (\neg p \rightarrow p)$

c) $(p \wedge q) \vee (\neg p) \vee (\neg q)$

d) $p \rightarrow \neg p$